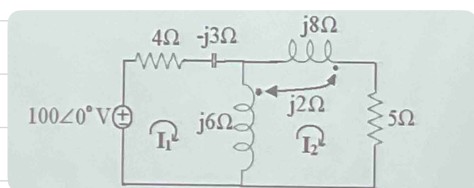


例 13-2



对网孔1:

$$-100 + I_1(4 - j3) + (-I_2 + I_1)(j6) - j2I_1 = 0$$

对网孔2:

$$(I_2 - I_1)(j6) + I_2(j8 + 5) + j2I_2 + j2(I_2 - I_1) = 0$$

可解得:

$$\begin{cases} I_1 = 20.3 \angle 3.5^\circ \text{ A} \\ I_2 = 8.693 \angle 19^\circ \text{ A} \end{cases}$$

例 13-2

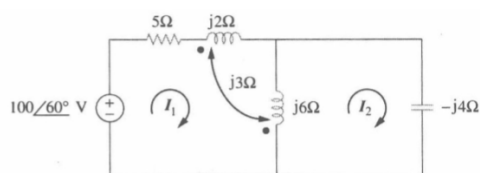


图 13-13 练习 13-2 图

对网孔1:

$$-100 \angle 60^\circ + (5 + j8 - 2 \times j2) \times I_1 - j6 \times I_2 + j2 \times I_2 = 0$$

对网孔2:

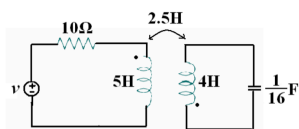
$$(I_2 - I_1) \times j6 + j2 \times I_1 - j4 \times I_2 = 0$$

可解得:

$$\begin{cases} I_1 = 17.889 \angle 86.57^\circ \text{ A} \\ I_2 = 26.83 \angle 86.57^\circ \text{ A} \end{cases}$$

例3:

对于图示电路, 计算其耦合系数; 并计算当 $v = 60 \cos(4t + 30^\circ) \text{ V}$ 时, 耦合电感器在 $t = 1\text{s}$ 时储存的能量。



$$L_1 = 5 \text{ H}, L_2 = 4 \text{ H}, M = 2.5 \text{ H}$$

则 $k = \frac{M}{\sqrt{L_1 L_2}}$, 代入得 $k = \frac{2.5}{\sqrt{5 \times 4}} = 0.56 > 0.5$

可得网孔电流方程:

$$\begin{cases} (10 + j20) I_1 + j10 I_2 = 60 \angle 30^\circ \\ j10 I_1 + (j16 - j4) I_2 = 0 \end{cases} \text{ 可得: } \begin{cases} I_1 = 3.905 \angle -19.4^\circ \\ I_2 = 3.254 \angle -160.6^\circ \end{cases}$$

转为时域有 $i_1 = 3.905 \cos(4t - 19.4^\circ)$, $i_2 = 3.254 \cos(4t - 160.6^\circ)$

$t = 1\text{s}$ 时, $4t = 4\pi = 4 \text{ rad} = 229.2^\circ$,

则 $i_1 = 3.905 \cos(229.2^\circ - 19.4^\circ) = -3.389 \text{ A}$

$i_2 = 3.254 \cos(229.2^\circ + 160.6^\circ) = 2.824 \text{ A}$

则存储的总能量为:

$$W = \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 + M i_1 i_2 = 20.73 \text{ J}$$